Analytical AI in Sustainability Communication Research and Education

Topic: ***The Adoption of AI in Strategic Communications*; Subtopic: *State of the Art: Empirical studies***

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Abstract

While *generative* AI tools have rapidly transformed communication workflows in recent years, the integration of artificial intelligence into strategic communication also builds on a longer trajectory of *analytical* AI. These algorithmic techniques—such as natural language processing (NLP) and computer vision—offer powerful capabilities for empirically analyzing sustainability communication at scale. This chapter argues for a more balanced understanding of AI adoption in the communication field, one that integrates the practical relevance of generative models with the empirical rigor and interpretive potential of analytical AI.

Focusing on sustainability communication, the chapter presents empirical evidence from recent content analyses that leverage transformer-based language models such as BERT. In one study, we use semantic similarity metrics derived from sentence embeddings to examine how key sustainability concepts—such as "energy" and "innovation" in Shell's messaging, and "conservation" and "duty" in WWF's campaigns—are framed across organizational communication materials. Our findings show that Shell's content emphasizes regulatory-economic terms and frames sustainability as a business opportunity, while WWF foregrounds community values and moral imperatives, aligning with a problem-oriented, duty-based frame. These results illustrate how analytical AI can extend existing theories in strategic communication by operationalizing abstract constructs such as framing, organizational identity, and value propositions.

In addition to its empirical contribution, the chapter reflects on the educational implications of AI in strategic communication. As communicative work becomes increasingly mediated by algorithmic tools, educational programs must evolve accordingly. The professional role of the communicator is shifting—from content producer to *content analyst*, from copywriter to *data interpreter*. We propose a pedagogical approach that equips students with both technical literacy and critical reflection, enabling them to use AI tools not only to automate

tasks but to conduct meaningful analysis, support evidence-based strategy, and critically evaluate communicative impact.

We argue that this evolving role of communication professionals must be underpinned by normative principles from the open science movement—transparency, reproducibility, and accessibility. Integrating these principles in AI-assisted research supports public accountability, fosters interdisciplinary collaboration, and prepares communicators to navigate the ethical challenges posed by algorithmic mediation.

Overall, this chapter contributes to the state of the art by showing how analytical AI methods can be integrated into strategic communication research and education. It bridges computational techniques with conceptual frameworks in sustainability communication, offering a template for combining empirical insight with professional relevance. It also emphasizes that the adoption of AI in communication is not only a technological shift, but a cognitive and institutional one—demanding new forms of literacy, pedagogy, and critical awareness among both researchers and practitioners.

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